A CASE-STUDY BASED REPORT

ON

Product Demand Forecasting Using LSTM and Gradient Boosting

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

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by

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1. ABSTRACT

Accurate product demand forecasting is essential for effective inventory management, supply chain optimisation, and business decision-making. Traditional forecasting methods often struggle to capture complex patterns in sales data, especially when demand is influenced by seasonal trends, promotions, customer behaviour, and market fluctuations. This project presents a hybrid demand forecasting model that combines Long Short-Term Memory (LSTM) networks and Gradient Boosting techniques to improve prediction accuracy. LSTM, a specialised type of recurrent neural network, is highly effective in learning long-term temporal dependencies from historical sales data, making it suitable for time-series forecasting. Gradient Boosting complements the LSTM model by capturing non-linear relationships between demand and additional influencing factors such as product category, pricing, promotional activities, holidays, and external variables.

The proposed approach involves collecting and preprocessing historical sales data, handling missing values, normalising features, and engineering relevant input variables. The LSTM model is trained to identify sequential patterns in demand over time, while the Gradient Boosting model learns feature-based relationships that contribute to demand variation. The predictions generated by both models are integrated to produce a more robust and reliable demand forecast. Model performance is evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

The hybrid forecasting framework aims to reduce forecasting errors, minimise stock shortages and overstock situations, and enhance inventory planning. By providing accurate demand estimates, businesses can optimise procurement, improve warehouse utilisation, reduce operational costs, and increase customer satisfaction. The proposed system is scalable and can be applied across various retail and e-commerce environments where demand patterns are dynamic and data-driven decision-making is critical.

2. INTRODUCTION

2.1 Background

Product demand forecasting is the process of predicting future customer demand for products. Accurate forecasting enables organisations to maintain the right inventory levels, reduce costs, avoid stock shortages, and improve customer satisfaction. With the rapid growth of retail and e-commerce, businesses require intelligent forecasting systems to make better decisions and improve supply chain efficiency.

Traditional statistical methods such as moving averages and exponential smoothing often struggle with complex sales patterns influenced by seasonality, promotions, pricing, and customer behaviour. Machine learning and deep learning techniques learn such patterns directly from historical data. Long Short-Term Memory (LSTM) networks are effective for analysing time-series data, while Gradient Boosting captures non-linear relationships between demand-related features.

2.2 Motivation for the Project

Inaccurate forecasts lead to two costly outcomes: stock-outs, which cause lost sales and unhappy customers, and overstocking, which ties up capital and increases storage and wastage costs. No single model is best at capturing both the sequential (time-dependent) behaviour of demand and the effect of external factors such as price, promotions and holidays. This motivates a hybrid approach that uses the strengths of both an LSTM and a Gradient Boosting model.

2.3 Overview of the Work Carried Out

The work carried out in this project consists of the following stages:

- •Collection of historical sales data in CSV format and loading it with Pandas.
- •Data preprocessing: handling missing values, removing duplicates, and normalising numerical features.
- •Feature engineering: calendar features, lag features, rolling averages, price, promotion and holiday indicators.
- •Training an LSTM model on sequences of past demand to learn temporal dependencies.
- •Training a Gradient Boosting model on engineered features to learn non-linear feature relationships.
- •Integrating (blending) the predictions of both models to obtain the final hybrid forecast.
- •Evaluating all models using MAE, RMSE and MAPE and comparing the hybrid model with the individual models.

2.4 Scope and Responsible-Use Statement

The scope of this project is short-term product-level demand forecasting for retail and e-commerce settings using historical sales data. The system is intended as a decision-support tool for inventory planning and not as a replacement for human judgement. Forecasts depend on the quality and representativeness of the data and may be less reliable during unprecedented events such as sudden market shocks. The data used is non-personal sales data, and no individual customer information is used or exposed.

3. PROBLEM STATEMENT

3.1 Case-Study Scenario

A retail and e-commerce company sells a large number of products whose demand changes with seasons, festivals, discounts and market conditions. The company stores daily sales records along with product category, selling price, promotion status and holiday information. The inventory team currently relies on simple averages to decide how much stock to order, which results in frequent stock shortages for fast-moving items and excess stock for slow-moving items. The company wants a data-driven forecasting system that predicts future demand more accurately.

3.2 Central Case-Study Question

Can a hybrid model that combines an LSTM network (for temporal patterns) and Gradient Boosting (for feature-based non-linear relationships) forecast product demand more accurately than either model used alone?

3.3 Formal Problem Definition

Given a historical time series of demand y(1), y(2), ..., y(t) and a set of exogenous features X(t) (price, promotion, holiday, category and calendar variables), the task is to learn a function f such that the demand for the next period is predicted as:

$$\hat{y}(t+1) = f(y(t-k+1 \dots t), X(t+1))$$

where k is the look-back window. In the hybrid framework, the final forecast is obtained as a weighted combination of both models:

$$\hat{y}_hybrid = w \cdot \hat{y}_LSTM + (1 - w) \cdot \hat{y}_GB, \quad 0 \leq w \leq 1$$

The objective is to minimise forecasting error measured by MAE, RMSE and MAPE.

3.4 Objectives that Guide the Study

- To preprocess historical sales data and engineer relevant time-series and business features.
- To build an LSTM model that learns long-term temporal dependencies in demand.
- To build a Gradient Boosting model that captures non-linear effects of price, promotions, holidays and category.
- To integrate both models into a hybrid forecasting framework.
- To evaluate and compare the models using MAE, RMSE and MAPE.
- To show how better forecasts can reduce stock shortages and overstock situations.

3.5 Assumptions and Limitations

- Past demand patterns are assumed to be informative about future demand.
- The dataset is assumed to be reasonably clean after preprocessing, with missing values handled appropriately.
- Information such as promotions and holidays is assumed to be known in advance for the forecast period.
- The model may not perform well for new products with little or no sales history (cold-start problem).
- Sudden external events (supply disruptions, competitor actions) are not explicitly modelled.
- Deep learning models require more training time and tuning than purely statistical methods.

4. DATASET AND EXPLORATORY DATA ANALYSIS

4.1 About the Dataset

The project uses a historical product sales dataset stored as a CSV file (.csv) and loaded with Pandas. Each record represents the sales of a product on a given date. The main attributes are described below.

Feature	Description	Type
date	Date of the sales record	Datetime
product_id / category	Identifier and category of the product	Categorical
price	Selling price of the product	Numeric
promotion	Whether a promotion/discount was active (0/1)	Binary
holiday	Whether the date is a holiday/festival (0/1)	Binary
units_sold (target)	Number of units sold, i.e. the demand to be forecast	Numeric

Note: the exact column names and number of records should be updated to match the CSV file finally used in the project.

4.2 Purpose of EDA

Exploratory Data Analysis (EDA) is carried out to understand the structure and quality of the data before modelling. It helps to identify trends, seasonality, outliers, missing values and relationships between demand and other factors, which in turn guides feature engineering and the choice of model parameters.

4.3 Preparation Steps Actually Performed

- **Data loading:** Loaded the CSV file and inspected shape, data types and summary statistics.
- **Missing values:** Filled or interpolated missing values and removed duplicate records.
- **Date handling:** Converted the date column to datetime format and sorted records chronologically.
- **Feature engineering:** Created day-of-week, month, week-of-year, lag (e.g. t-1, t-7) and rolling-mean features.
- **Normalisation:** Scaled numerical features using Min-Max scaling (required for the LSTM).
- **Train-test split:** Split the data chronologically into training and testing sets (e.g. 80% / 20%) to avoid data leakage.

4.4 EDA Exploration - Implementation

The following Python code (Pandas, NumPy, Matplotlib) outlines the EDA performed:

```
import pandas as pd, numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('sales_data.csv', parse_dates=['date'])
df = df.sort_values('date')
```

```
print(df.shape); print(df.info()); print(df.isnull().sum())
print(df.describe())

# Demand over time
df.groupby('date')['units_sold'].sum().plot(figsize=(10,4))
plt.title('Total Demand Over Time'); plt.show()

# Effect of promotion and day of week
df.groupby('promotion')['units_sold'].mean().plot(kind='bar')
df.groupby(df['date'].dt.dayofweek)['units_sold'].mean().plot(kind='bar')

# Correlation heatmap
print(df.select_dtypes('number').corr())
```

Suggested screenshots to paste here from your notebook: (a) dataset head and info, (b) demand-over-time line plot, (c) promotion/holiday vs. demand bar charts, (d) correlation heatmap.

4.5 Summary of EDA Findings

- •Demand shows a clear time-dependent pattern with trend and seasonal variation, which supports the use of an LSTM.
- •Promotions and holidays are expected to raise average demand, and price is expected to be negatively related to units sold; such feature effects are well captured by Gradient Boosting.
- •Lag and rolling-average features are strongly related to current demand, confirming temporal dependence.
- •Outliers and missing values were treated during preprocessing so that they do not distort model training.

Note: replace these statements with the actual observations and values from your own EDA output.

5. ALGORITHM SELECTION AND IMPLEMENTATION

5.1 Long Short-Term Memory (LSTM)

LSTM is a special type of recurrent neural network that overcomes the vanishing-gradient problem of standard RNNs using a memory cell controlled by three gates: the forget gate, input gate and output gate. This lets the network remember important information over long periods, which makes it well suited to sales time series with trend and seasonality. In this project, windows of the previous k days of demand are given as input and the model predicts the next day's demand.

5.2 Gradient Boosting

Gradient Boosting is an ensemble technique that builds decision trees sequentially, where each new tree is trained to correct the residual errors of the previous trees by following the negative gradient of the loss function. It handles non-linear relationships and feature interactions well, so it is used to model the effect of price, promotions, holidays, category and calendar features on demand.

5.3 Hybrid Model

The predictions of the LSTM and Gradient Boosting models are combined using a weighted average, where the weight w is chosen using validation data. This combines the temporal learning ability of the LSTM with the feature-learning ability of Gradient Boosting, giving a more robust forecast.

5.4 Overall Workflow

Historical Sales Data → Preprocessing → Feature Engineering → Scaling → Train–Test Split → Train LSTM + Train Gradient Boosting → Blend Predictions → Evaluate (MAE, RMSE, MAPE) → Demand Forecast

5.5 Implementation

```
from sklearn.preprocessing import MinMaxScaler
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout

# ---- LSTM ----
lstm = Sequential([
    LSTM(64, input shape=(k, 1)),
    Dropout(0.2),
    Dense(1)])
lstm.compile(optimizer='adam', loss='mse')
lstm.fit(X_seq_train, y_train, epochs=50, batch_size=32,
         validation_split=0.1)
lstm pred = lstm.predict(X seq test).ravel()

# ---- Gradient Boosting ----
gb = GradientBoostingRegressor(n_estimators=300, learning_rate=0.05,
                               max_depth=4, random_state=42)
gb.fit(X_feat_train, y_train)
gb_pred = gb.predict(X_feat_test)

# ---- Hybrid ----
w = 0.5
hybrid_pred = w*lstm_pred + (1-w)*gb_pred
```

Note: the Python library list in your abstract mentions Pandas, NumPy, Matplotlib and Scikit-learn; the LSTM code additionally needs TensorFlow/Keras, so add it to the software requirements.

5.6 Evaluation Metrics

Mean Absolute Error (MAE) = $(1/n)\ \Sigma\ |y - \hat{y}|$

Root Mean Square Error (RMSE) = $\sqrt{[\ (1/n)\ \Sigma\ (y - \hat{y})^2\ ]}$

Mean Absolute Percentage Error (MAPE) = $(100/n)\ \Sigma\ |(y - \hat{y})\ /\ y|$

MAE gives the average size of the error, RMSE penalises large errors more strongly, and MAPE expresses the error as a percentage of actual demand, making it easy to compare across products.

5.7 Results Comparison

Model	MAE	RMSE	MAPE (%)
LSTM	—	—	—
Gradient Boosting	—	—	—
Hybrid (LSTM + GB)	—	—	—

Note: fill this table with the actual values obtained after running your notebook.

6. CONCLUSION

This project presented a hybrid demand forecasting framework that combines LSTM networks and Gradient Boosting. The LSTM captures the sequential patterns in historical sales, while Gradient Boosting models the non-linear influence of price, promotions, holidays and other external factors. Integrating the two predictions produces a more robust and reliable forecast than relying on a single model, and is evaluated using MAE, RMSE and MAPE.

Accurate demand estimates help businesses minimise stock shortages and overstock situations, optimise procurement, improve warehouse utilisation, reduce operational costs and increase customer satisfaction. The system is scalable and can be applied across retail and e-commerce environments. Future work may include adding external data such as weather and competitor prices, trying advanced models such as XGBoost, LightGBM or Transformers, and deploying the model as a real-time forecasting dashboard.

Signature of the Team members